

FUSION HIGH EDUCATIONAL ECOSYSTEM

Complete Role-by-Role System Documentation Guide

Comprehensive Technical Architecture, Database Schemas & Portal-by-Portal Manual
Covering Limpopo (Capricorn South) & Gauteng (Tshwane Districts)

SYSTEM CHAPTERS & PORTAL BREAKDOWN

- Chapter 1: Executive Administrator Operations (Command Center, User Onboarding, Derbies, Marks, Timetables, Fees)
- Chapter 2: Educator & Teacher Workspace (Teaching Workload, Resources Upload Studio, CAPS Markbook, Attendance, AI Planner)
- Chapter 3: Parent & Family Learning Portal (Multi-Child Monitoring, Subject Reports, 20-min Consultations, Fee Invoices)
- Chapter 4: Learner & Student Learning Hub (Subject Topics, School-Aware AI Tutor, Fusion Arcade, NSFAS Bursaries, Careers)
- Chapter 5: Technical Infrastructure & Database Model (PostgreSQL Relational Schema, Security & DBE CAPS Compliance)

COMPLETE TABLE OF CONTENTS

- Multi-Tenant Architecture & Governance Hierarchy
- Chapter 1: Administrator Hub (Command Center, Users, Derbies, Moderation, Fees, Matric)
- Chapter 2: Educator Workspace (Workload, Resource Uploads, Markbook, AI Planner, Homework)
- Chapter 3: Parent Portal (Multi-Child Progress, 20-min Consultations, Fee Statements)
- Chapter 4: Learner Hub (Curriculum Topics, School-Aware AI Tutor, Fusion Arcade, NSFAS)
- Security, Role-Based Access Control (RBAC) & DBE CAPS Protocols

Chapter 1: Administrator Operations & Multi-School Governance

The Executive Administrator Dashboard provides oversight over all 12 institutions with real database telemetry.

- Multi-School Command Center: Comparative metrics across all schools in Limpopo and Gauteng.
- User Directory & SubAdmin Appointments: Assigns dedicated campus administrators and educators with automated credentials.
- Inter-School Derby League: Schedules competitive sports matches and academic Olympiads with live standings.
- CAPS Mark Audits: Continuous assessment moderation verifying DBE assessment protocols.
- Conflict-Free Timetables: Algorithmic master schedules allocating teachers and classrooms with zero clashes.
- Financial Management: Itemized term fee invoicing, digital payment receipts, and collection auditing.

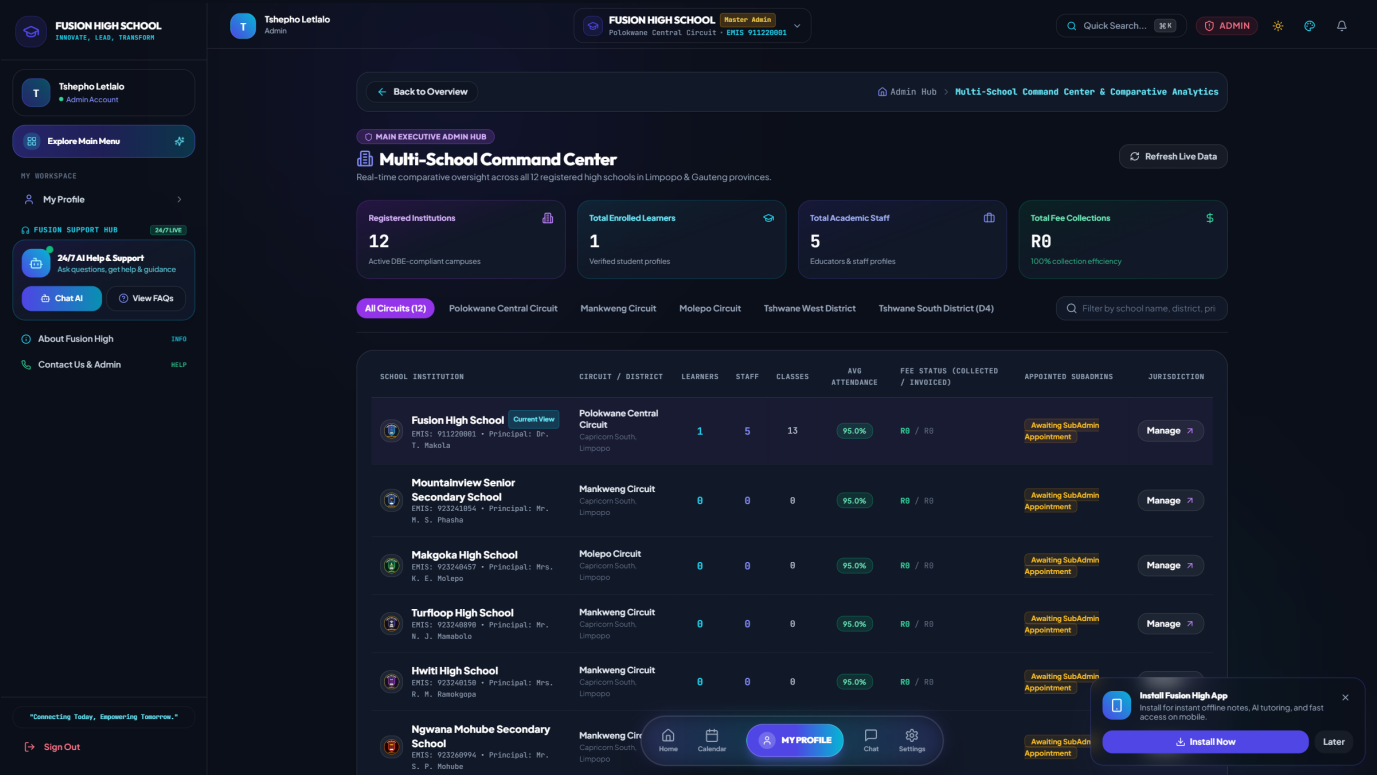


Figure 1: Multi-School Command Center displaying live telemetry across all 12 registered institutions.

Chapter 2: Educator & Teacher Workspace

The Educator Workspace is designed to streamline daily teaching activities and curriculum delivery.

- Teaching Schedule: Real-time timetable of current and upcoming class periods.
- Resource Upload Studio: Teachers upload past papers, exam memos, study guides, and worksheets on demand.
- CAPS Digital Markbook: Records test and assignment marks with automatic SBA weighting calculations.
- Attendance Register: Period-by-period attendance with automated parent notifications.
- AI Lesson Planner: Generates 45-minute CAPS-aligned teaching plans and examination question papers.

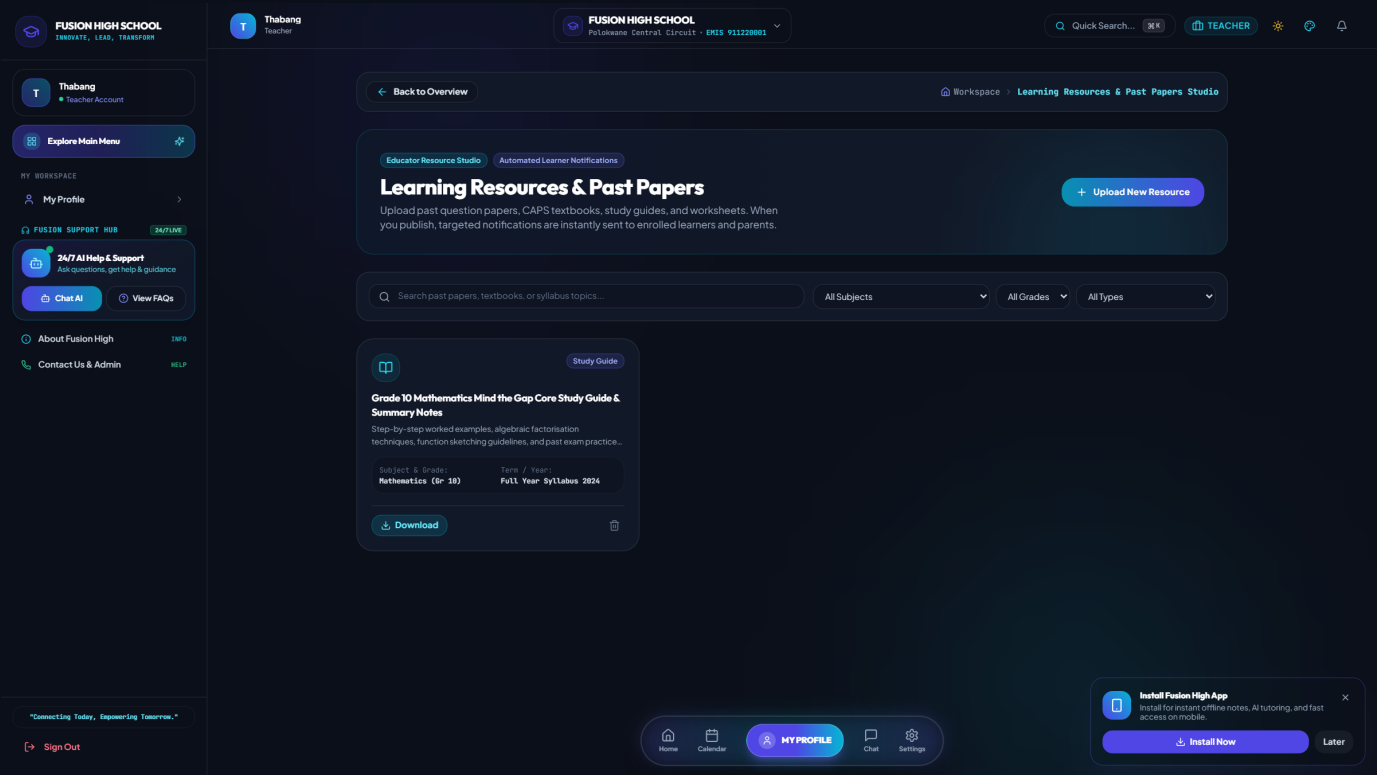


Figure 2: Learning Resources & Past Papers Studio allowing educators to upload materials autonomously.

Chapter 3: Parent & Family Learning Portal

The Parent Portal bridges home and school communication with real-time academic transparency.

- Multi-Child Dashboard: Monitor all enrolled children across grades from a single parent account.
- Academic Reports: Live subject progress bars, task marks, and official signed CAPS report cards.
- 20-Minute Consultation Scheduler: Parents select their child and educator to book open time slots.
- Financial Transparency: Digital fee statements, itemized invoices, and automated Sunday digests.
- Daily Attendance & Timetable: Tracks period attendance and classroom schedules.

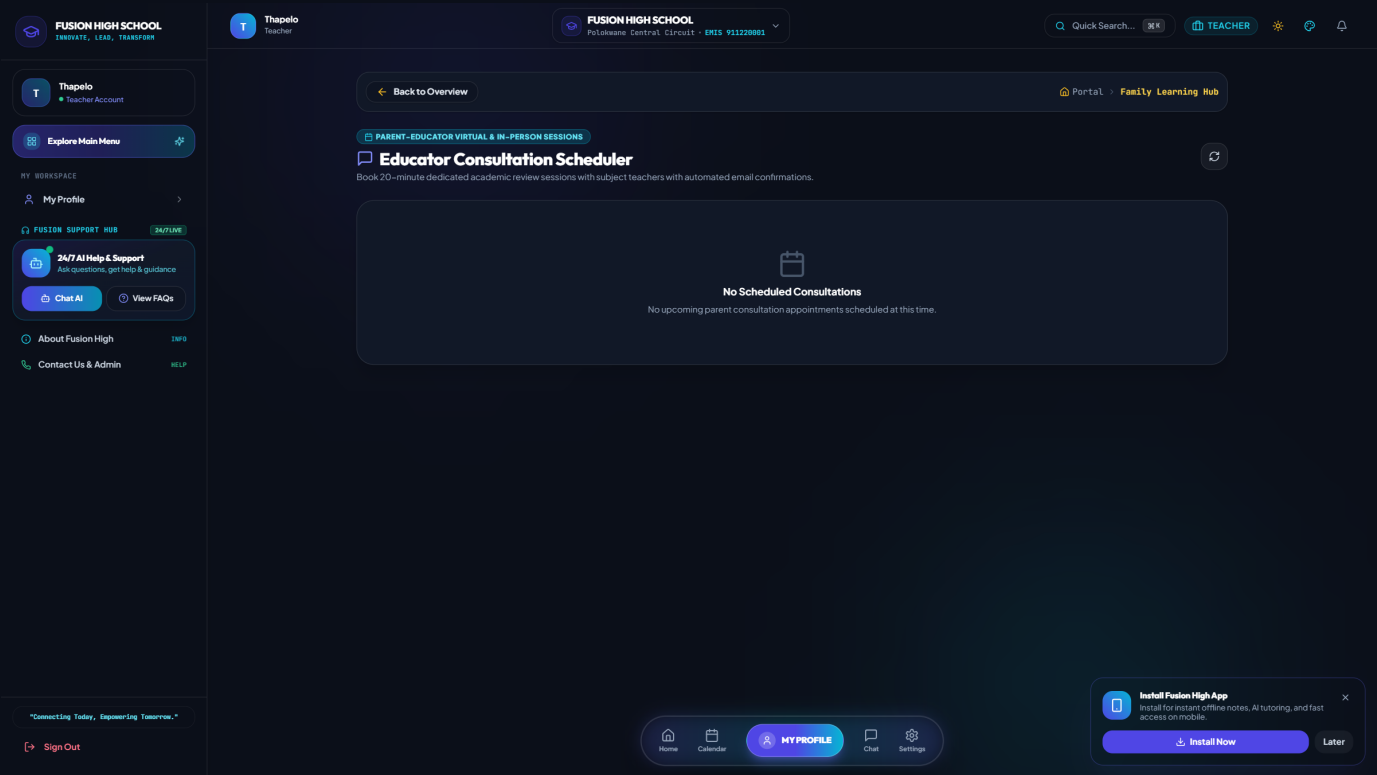


Figure 3: Parent-Educator 20-Minute Consultation Scheduler with automated email confirmations.

Chapter 4: Learner & Student Learning Hub

The Student Learning Hub motivates learners through structured curriculum support and interactive revision.

- School-Aware AI Tutor: Multi-tenant grounding sensing the learner's campus with zero cross-school leakage.
- Subject Curriculum & Notes: Step-by-step topic guides and formula summaries across all CAPS subjects.
- Fusion Arcade Games: Gamified quizzes awarding XP, streak multipliers, and leaderboard rankings.
- NSFAS & Tertiary Bursaries: Degree matching engine evaluating simulated APS points.
- Career Advisor: Guides learners on faculty requirements for South African universities.

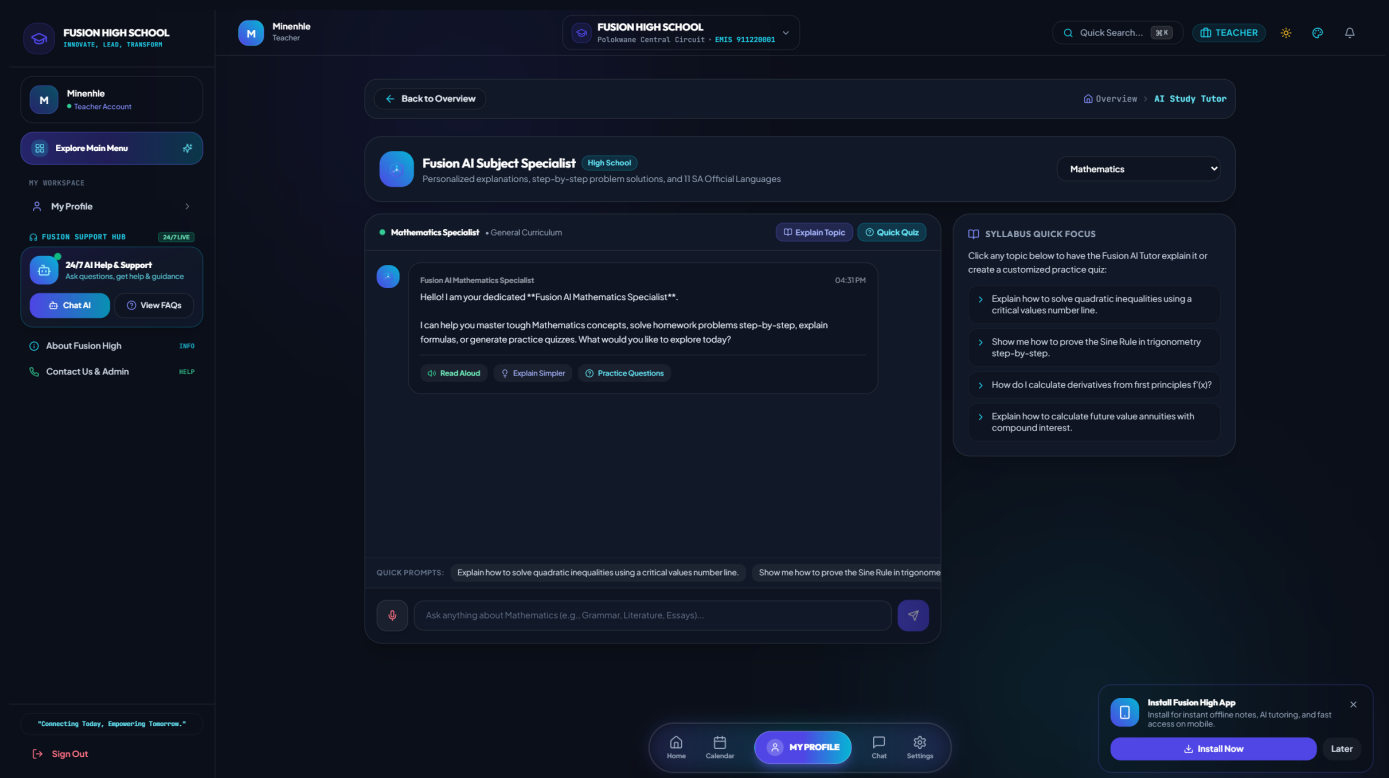


Figure 4: School-Aware AI Academic Tutor providing bilingual CAPS revision.